M01 - Data Foundations + Essential Excel

15-lesson beginner-sufficient learner book - RC1

How to use each lesson

Read the explanation, reproduce the small example, complete the matching guided lab, then close the book for the independent task. A lesson is not mastered merely because the example works.

DE01-01 - What data is: tables, records, fields and data types

Why this matters	You need a precise mental model before touching formulas or SQL.

Core explanation

• A row/record represents one observation at the table grain.

• A column/field stores one attribute consistently across rows.

• Types matter: text, numbers, dates and booleans behave differently.

• A cell can look numeric while actually being stored as text.

Concrete example

In an orders table, OrderID is an identifier, OrderDate is a date, Quantity is numeric, and Region is categorical text.

Common failure	Do not treat formatting as data type. A date-looking string may still be text.
Proof before move-on	Given a 6-column table, state the row grain and classify every column type.

Explain aloud

In 60-90 seconds explain what data is: tables, records, fields and data types, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-02 - Grain: what one row represents

Why this matters	Wrong grain causes duplicate counts, wrong joins and bad aggregates.

Core explanation

• Write a grain sentence before calculating anything.

• Different tables can describe the same business at different grains.

• Order header, order line, payment and refund event are different grains.

• Aggregation changes grain; filtering usually does not.

Concrete example

```
If one order has three line items, an order-line table has three rows for that order. COUNT(rows) is not automatically COUNT(or
```

Common failure	Never say "the grain is customer" if multiple rows per customer are expected.
Proof before move-on	Write grain sentences for Raw_Sales, Customer_Lookup and a refund-event table.

Explain aloud

In 60-90 seconds explain grain: what one row represents, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-03 - Keys, duplicates, NULLs and missing data

Why this matters	You must distinguish legitimate repetition from bad duplication.

Core explanation

• A primary/business key should uniquely identify the intended grain.

• Repeated customer IDs can be valid in order data.

• Exact duplicate rows may indicate ingestion or copy errors.

• Blank, NULL and zero do not mean the same thing.

Concrete example

Customer C100 may appear in five order rows legitimately. Two identical order-line rows with the same line key are suspicious.

Common failure	Do not delete repeats until you prove they are duplicates at the table grain.
Proof before move-on	Identify the candidate key, expected repeats and suspicious duplicates in a sample.

Explain aloud

In 60-90 seconds explain keys, duplicates, nulls and missing data, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-04 - Data quality dimensions and validation thinking

Why this matters	Engineering starts with trust in inputs, not with attractive outputs.

Core explanation

• Completeness checks required fields.

• Uniqueness checks keys, not every repeated value.

• Validity checks domains, ranges and data types.

• Consistency checks labels/relationships across sources.

• Timeliness checks whether data is current enough for the decision.

Concrete example

A negative quantity may be invalid, or it may represent a return depending on the source contract. Investigate before "fixing" it.

Common failure	Do not use generic cleaning rules without business meaning.
Proof before move-on	Design five checks for order data and state what each failure would mean.

Explain aloud

In 60-90 seconds explain data quality dimensions and validation thinking, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-05 - Meet Excel: workbook, worksheet, cells and formulas

Why this matters	Excel is used as a practical grid for inspecting and validating small datasets.

Core explanation

• Workbook = file; worksheet = tab.

• Cell addresses combine column letter and row number.

• Formula bar shows the stored formula/value.

• Name box and sheet tabs help navigation.

Concrete example

Navigate to Raw_Sales!F14, then to Formula_Practice!D4. Explain what the address means.

Common failure	Do not confuse displayed value with underlying formula.
Proof before move-on	Navigate to three named cells without following a video.

Explain aloud

In 60-90 seconds explain meet excel: workbook, worksheet, cells and formulas, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-06 - Enter, save and structure tabular data safely

Why this matters	Good structure prevents accidental analysis errors.

Core explanation

• Keep one untouched raw copy.

• Use one header row with unique column names.

• Avoid merged cells and blank separator rows inside data.

• One column should represent one field with one meaning.

Concrete example

```
Create a working copy before editing raw sales. Put notes outside the data table, not inside random rows.
```

Common failure	Do not sort a single column independently from its row.
Proof before move-on	Diagnose at least four structural problems in a messy sheet.

Explain aloud

In 60-90 seconds explain enter, save and structure tabular data safely, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-07 - Sort and filter without damaging data

Why this matters	Sort/filter are inspection tools; careless use can break row relationships.

Core explanation

• Filter hides rows; it does not delete them.

• Sort the whole table/range, not one isolated column.

• Clear filters before concluding that rows are missing.

• Check row count before and after operations.

Concrete example

Filter Region = East and Status = Completed; record visible row count, then clear filters and verify total row count returns.

Common failure	Do not copy filtered visible rows back over the raw source.
Proof before move-on	Complete a filter/sort inspection and prove row integrity.

Explain aloud

In 60-90 seconds explain sort and filter without damaging data, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-08 - Basic formulas and essential functions

Why this matters	Formulas turn explicit inputs into reproducible calculations.

Core explanation

• Start formulas with =.

• Use cell references instead of hard-coded numbers when inputs already exist.

• SUM, AVERAGE, MIN, MAX and COUNT solve common summaries.

• Check at least one result manually.

Concrete example

Revenue in G2 can be =E2*F2. Total revenue can use SUM(G2:G20).

Common failure	A formula returning a number is not proof it is correct.
Proof before move-on	Build revenue and four summary formulas from blank cells.

Explain aloud

In 60-90 seconds explain basic formulas and essential functions, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-09 - Relative and absolute references

Why this matters	Copied formulas must move only the references that should move.

Core explanation

• Relative A2 changes when copied.

• Absolute H2 stays fixed.

• Mixed references lock one dimension.

• Choose based on business meaning, not keyboard habit.

Concrete example

```
If every row uses the same tax rate in H2, =G5*(1+$H$2) should keep H2 fixed when filled down.
```

Common failure	Do not use absolute references everywhere; that can silently repeat the wrong input.
Proof before move-on	Predict three copied formulas before filling them.

Explain aloud

In 60-90 seconds explain relative and absolute references, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-10 - Excel Tables and structured data habits

Why this matters	Tables make small datasets easier to filter, expand and reference safely.

Core explanation

• A Table is an Excel object, not just colored cells.

• Headers remain explicit and filters are built in.

• Calculated columns can propagate formulas.

• Tables do not fix bad grain or duplicate keys automatically.

Concrete example

Convert a clean working range into a Table and confirm the header/filter behavior.

Common failure	Do not assume "Table" means the data is trustworthy.
Proof before move-on	Create a Table, add one row and verify the structure expands correctly.

Explain aloud

In 60-90 seconds explain excel tables and structured data habits, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-11 - Practical cleaning and lookup awareness

Why this matters	You need to standardize obvious issues while preserving evidence.

Core explanation

• TRIM/CLEAN-like operations address whitespace/control characters.

• Case standardization can reduce label inconsistency.

• Lookups require a trustworthy key on the lookup side.

• XLOOKUP is useful but should not hide duplicate keys.

Concrete example

Before looking up customer segment, confirm CustomerID is unique in Customer_Lookup.

Common failure	Do not use a lookup result to cover a duplicate-key problem.
Proof before move-on	Clean one label field and perform or explain one key-based lookup.

Explain aloud

In 60-90 seconds explain practical cleaning and lookup awareness, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-12 - PivotTable and Power Query awareness

Why this matters	These tools appear frequently in analyst and data-prep workflows, even though deep mastery comes later.

Core explanation

• PivotTables summarize data quickly by dimensions/measures.

• Power Query records repeatable transformation steps.

• Both still depend on correct grain and source quality.

• This module uses awareness, not advanced modeling.

Concrete example

Build a simple Region x Revenue PivotTable or explain the field placement if your Excel version differs.

Common failure	Do not mistake drag-and-drop speed for validated analysis.
Proof before move-on	Create one PivotTable summary and state one validation check.

Explain aloud

In 60-90 seconds explain pivottable and power query awareness, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-13 - Chart choice and basic visual communication

Why this matters	A correct number can still be miscommunicated by a poor chart.

Core explanation

• Use bars/columns for category comparison.

• Use lines for ordered time trends.

• Start from a clear question and metric.

• Label axes/units and avoid decorative clutter.

Concrete example

Compare revenue by region with a column chart; use a line chart only if the x-axis is ordered time.

Common failure	Do not use a pie chart for many categories or a line chart for unordered labels.
Proof before move-on	Choose and justify one chart for a supplied business question.

Explain aloud

In 60-90 seconds explain chart choice and basic visual communication, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-14 - Reconciliation, sanity checks and source preservation

Why this matters	Professionals prove the result, they do not just produce it.

Core explanation

• Record source row count and key totals before transformation.

• Reconcile calculated totals to an independent method where possible.

• Check min/max and suspicious outliers.

• Preserve raw data and document assumptions.

Concrete example

```
If SUM(Revenue) is 24,500, manually verify a small sample and compare grouped totals back to the grand total.
```

Common failure	Do not accept a dashboard total just because it looks plausible.
Proof before move-on	Produce a short validation log with row count, total, duplicates and missingness.

Explain aloud

In 60-90 seconds explain reconciliation, sanity checks and source preservation, the input/output grain or program state, and one way a plausible-looking result could still be wrong.

DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain -> save evidence

Why this matters	This lesson joins the module into one repeatable working habit.

Core explanation

• Read the business question before choosing tools.

• State grain and checks.

• Do the smallest correct calculation.

• Validate, explain limitations and save evidence.

Concrete example

For a regional sales question: inspect grain, compute revenue, summarize by region, reconcile to total, create one chart, and save a note explaining the result.

Common failure	Do not jump to charts before defining the metric and validating the data.
Proof before move-on	Complete the module practical without the lesson book open.

Explain aloud

In 60-90 seconds explain mini workflow: inspect -> calculate -> validate -> explain -> save evidence, the input/output grain or program state, and one way a plausible-looking result could still be wrong.